NEGF Modelling of RTD

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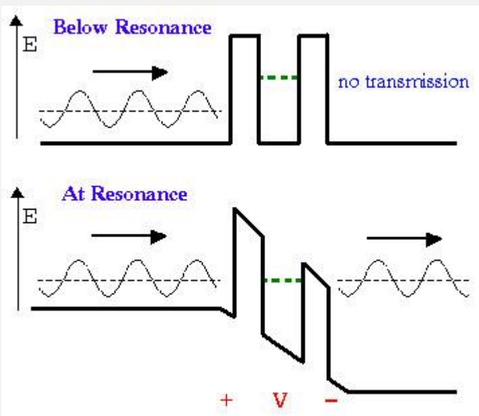
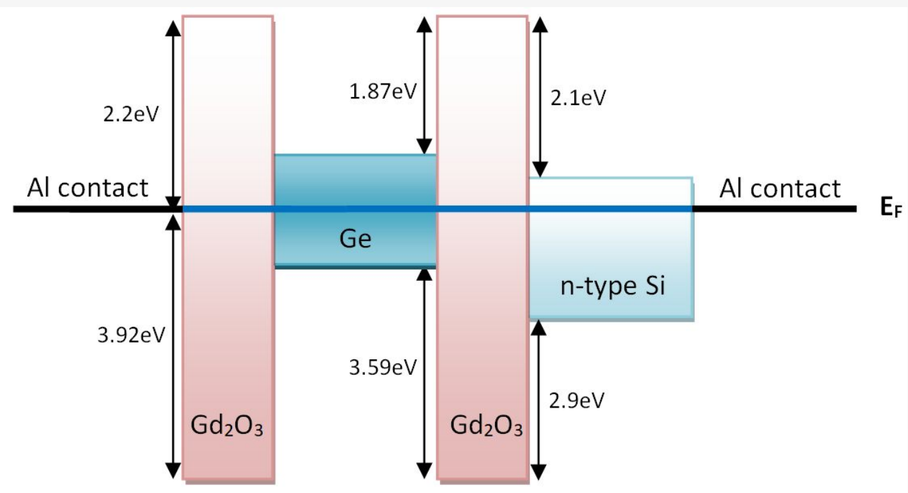
Resonant Tunneling Diode (RTD)

Resonant tunnelling diode under scrutiny here is a three layer stack of a semiconductor (Ge) sandwiched between two insulator (Gd_2O_3) layers. Transport behaviour of an RTD is governed majorly by two quantum phenomenon: **discretization of energy levels in a quantum well and electron tunneling across a thin barrier.**

The region between the two barriers acts as a quantum well with discrete energy states. Spacing between the energy levels varies inversely as the square of length of the well. Resonant tunneling through the double-barrier structure occurs when the energy of the electrons flowing from the source (left) coincides with one of the discrete energy levels in the well. Thus, on varying the longitudinal energy level of incident electrons, we expect to see spikes in the transmission coefficient close to the well eigenvalues. On the other hand, position of the bounded states within the well can be modulated with a gate bias . This results in what can be seen in the I-V characteristics as peak current and valley current. The region with the reducing current is known as a **negative differential resistance** region.



Source: https://nanohub.org/wiki/RTD



Schematic of the RTD barrier under study. Note: Image not to scale

The 1985 paper by Tsuchiya et. al. documented qualitative behaviour of a AlAs/GaAs/AlAs RTD in terms of the transmission coefficient and I-V characteristics. Our goal here has been to reproduce these results for a **RTD (Gd_2O_3/Ge/Gd_2O_3)** being fabricated at CEN, IIT Bombay.

NEGF as a computational tool

Simulation of a nanoscale semiconductor device for transport characteristics involves solving the Schrödinger equation. Computation involves solving the Schrödinger equation for a discrete set of points along the device. In discrete space, SE reduces to an LCCDE. A standard trick for solving LCCDE is taking it to Fourier domain which converts time-domain convolutions to frequency (or energy, for waves) domain multiplications. Non-equilibrium Green's Function turns out to be the impulse response of the Schrödinger equation.

Modified Schrödinger equation for an open quantum system is given by

$$i\frac{h}{2\pi}\frac{\partial\psi(x,t)}{\partial t} = [H+\Sigma]\psi + S$$

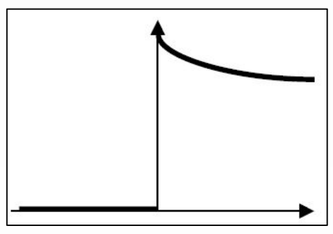
where S is the source term representing inflow while Σ represents the contact coupling and forms the outflow term. Taking the Fourier transform, to enter energy domain, we get

$$E\Psi = [H+\Sigma]\Psi + S$$

$$\Psi = \frac{S}{[EI - H - \Sigma]}$$

To get the impulse response, we substitute an identity matrix for the source term S and hence, get the expression for retarded Green's function

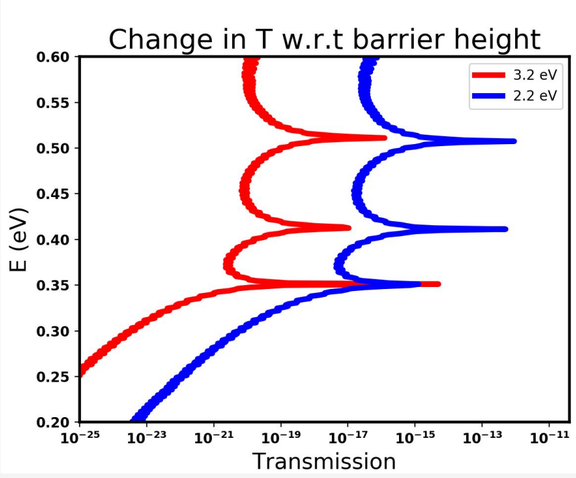
$$G^R(E) = [EI - H - \Sigma]^{-1}$$



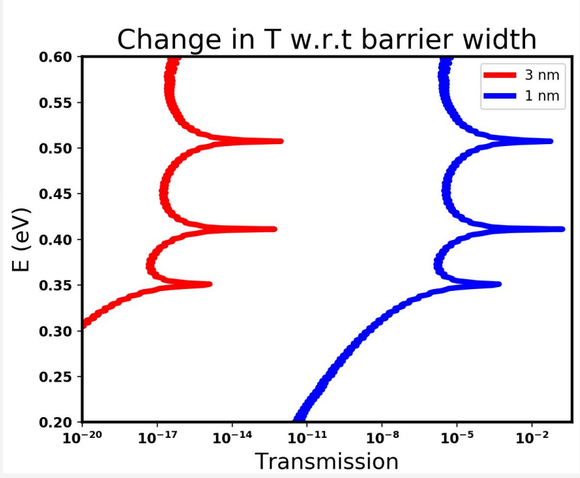
Time impulse response for diagonal terms of the retarded Green's function

Simulation Results

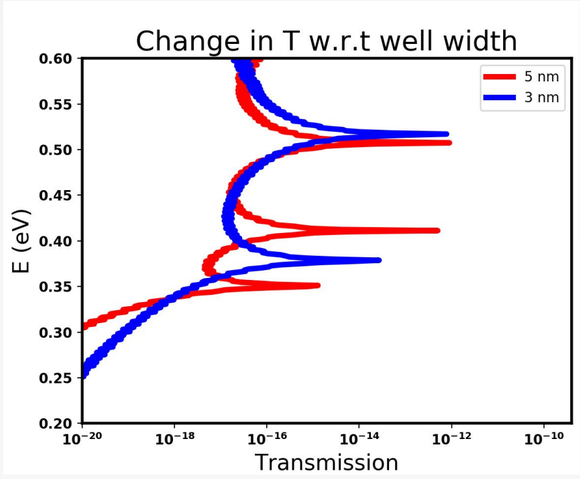
The device was simulated on Python to observe its transmission and I-V characteristics. We have made a flatband approximation. Results have been compared for varying barrier height, barrier width and well width.



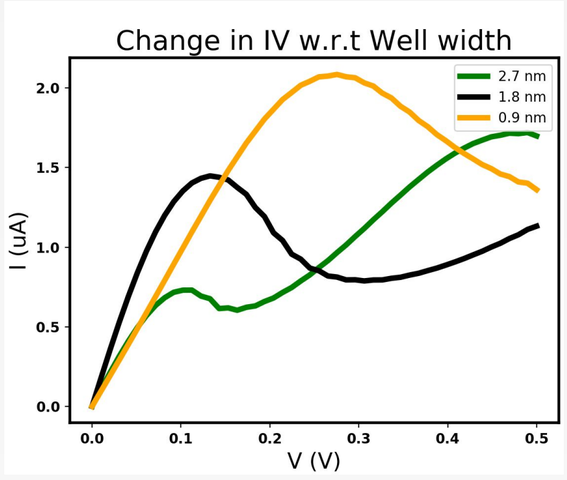
Increase in insulator barrier height clearly reduces the transmission coefficient. The spikes coincide since the energy eigenvalues of the well remain unchanged.



Increase in insulator thickness reduces the transmission coefficient. The spikes coincide since the energy eigenvalues of the well remain unchanged.



A smaller well width gives larger energy eigenvalues. This is reflected in the transmission coefficient spikes



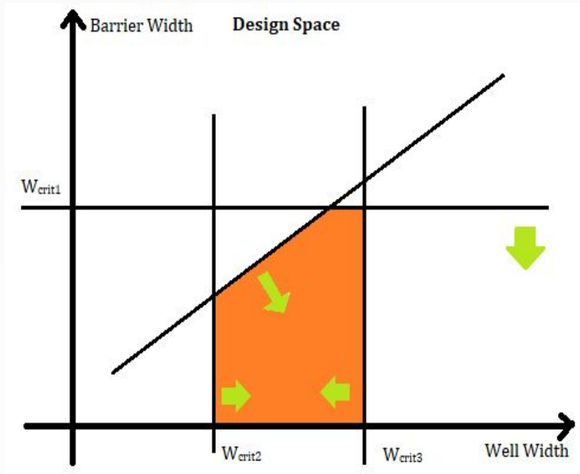
I-V plots for a simple RTD model, exhibiting negative resistance. Voltage corresponding to peak current shifts left with increasing well width.

Conclusions

1. NEGF can be used to model a RTD stack for its transmission coefficient and I-V characteristics.
2. Transmission coefficient decreases with increasing barrier height and barrier width. Transmission coefficient values are very small for practical use with any barrier thickness above 1nm.
3. RTD exhibits a negative resistance region. The voltage for peak current reduces with increasing well width. Changing barrier parameters only affect the peak current values but do not shift the I-V laterally.
4. I-V can be brought close to experimental observations by including incoherent transport formalism and lifting the linear potential drop assumption by self-consistently solving the Poisson and Schrödinger equations.

Considering our simulation results, we can propose a design space for the device.

- Practical applications requires a considerable transmission coefficient thus $W_{bar} < W_{crit1}$.
- Low voltage operations requires a large well width, $W_{well} > W_{crit2}$.
- Wider negative resistance region implies small well width, $W_{well} < W_{crit3}$.
- A larger I-V slope for fast switching requires a small ratio between W_{bar} and W_{well}. Thus, $W_{bar} < kW_{well}$



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